

سؤال

Example 1 :

$$U = \frac{1}{\binom{n}{2}} \sum_{i < j} g(x_i, x_j) \quad x_1, \dots, x_n \stackrel{iid}{\sim} P$$

Assumption $\|g\|_\infty \leq b$.

$$\mathbb{P}(|U - \mathbb{E}(U)| \geq t)$$

$$\begin{aligned} & \text{(McDiarmid)} \quad \frac{-2t^2}{L_n^2} \\ & \leq 2e^{-\frac{2t^2}{L_n^2}} \\ & = 2e^{-\frac{2t^2}{n \cdot 16b^2/n^2}} \\ & = 2e^{-\frac{nt^2}{8b^2}} \end{aligned}$$

$$\begin{aligned} & f(x_1, \dots, x_{k-1}, x'_k, x_{k+1}, \dots, x_n) - \\ & f(x_1, \dots, x_k, \dots, x_n) \leq \frac{2b(n-1)}{\frac{n(n-1)}{2}} \\ & = \frac{4b}{n} \end{aligned}$$

Example: Rademacher Complexity

$$Z = \sup_{\underline{a} \in \underline{A} \subseteq \mathbb{R}^n} \langle \underline{a}, \underline{\epsilon} \rangle$$

$$\underline{\epsilon} = (\epsilon_1, \dots, \epsilon_n)$$

$$\epsilon_i \stackrel{iid}{=} \begin{cases} 1 & P = 1/2 \\ -1 & P = 1/2 \end{cases}$$

$$\mathbb{P}(|Z - \mathbb{E}(Z)| \geq t) \leq 2e^{-\frac{2t^2}{n(4a_{\max}^2)}} \quad \forall a \in A: |a_i| \leq a_{\max}$$

$$f(\underline{\epsilon}) = \sup_{\underline{a} \in A} \langle \underline{a}, \underline{\epsilon} \rangle \quad L \leq 2a_{\max}$$

$$\begin{aligned} \mathbb{P}(|Z - \mathbb{E}(Z)| \geq t) & \leq 2e^{-\frac{2t^2}{\sum L_i^2}} \quad |a_i| < a_{\max}^i \\ & \leq e^{-\frac{(2t^2)}{4 \sum_{i=1}^n (a_{\max}^i)^2}} \quad L_i \leq 2a_{\max}^i \end{aligned}$$

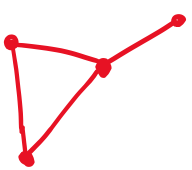
$$Z \sim \text{subG} \left(\sqrt{\sum_{i=1}^n (a_{\max}^i)^2} \right)$$

$$\mathbb{P}(|Z - \mathbb{E}(Z)| \geq t) \leq 2 e^{-\frac{t^2}{2 \left[\sum_{i=1}^n (a_{\max}^i)^2 / 4 \right]}} = 2 e^{-\frac{2t^2}{\sum_{i=1}^n (a_{\max}^i)^2}}$$

$$Z \sim \text{subG} \left(\sup_{a \in \mathcal{A}} \|a\| \right)$$

Example 3.

$C(G)$ clique # of graph G .

$$C \left(\text{graph} \right) = 3$$


G could be a random graph.

$$Z_{ij} \stackrel{\text{iid}}{\sim} \begin{cases} 1 & p \\ 0 & 1-p \end{cases}$$

$$\mathbb{P} \left(\frac{1}{n} |C(G) - \mathbb{E}(C(G))| \geq t \right) \leq 2 e^{-\frac{2 t^2 n^2}{n(n-1)}}$$

□

f L -Lipschitz on Norm-2

$$\exists L \forall \underline{x}, \underline{x}': |f(\underline{x}) - f(\underline{x}')| \leq L \|\underline{x} - \underline{x}'\|$$

$$X_1 \dots X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$$

$$\text{Theorem: } \mathbb{P}(|f(\underline{X}) - \mathbb{E}(f(\underline{X}))| \geq t) \leq 2 e^{-\frac{t^2}{2L^2}}$$

Proof: $\|\nabla f\|_2 \leq L.$

Lemma: $\phi(\cdot)$ convex
 $\underline{x}, \underline{y} \sim^{iid} \mathcal{N}(0, I)$

$$\mathbb{E} \left\{ \phi(f(\underline{x}) - \mathbb{E}(f(\underline{x}))) \right\}$$

$$\leq \mathbb{E} \left\{ \phi\left(\frac{\pi}{2} \langle \nabla f(\underline{x}), \underline{y} \rangle\right) \right\}$$

Take $\phi(t) = e^{\lambda t}$: $\mathbb{E} \left\{ e^{\lambda(f(\underline{x}) - \mathbb{E}(f(\underline{x})))} \right\}$

$$\leq \mathbb{E} \left(\phi\left(\frac{\pi}{2} \langle \nabla f(\underline{x}), \underline{y} \rangle\right) \right)$$

$$= \mathbb{E}_{\underline{x}} \mathbb{E}_{\substack{\underline{y} | \underline{x} \\ \underline{y}}} \left\{ \phi\left(\frac{\pi}{2} \langle \nabla f, \underline{y} \rangle\right) \right\}$$

$\sum_{\underline{y}} \mathcal{N}(\underline{y}^T \underline{0}, \frac{\underline{y}^T I \underline{y}}{\|\underline{y}\|^2})$

$$\leq \mathbb{E}_{\underline{x}} \left\{ e^{\frac{\pi^2}{4} \lambda^2 \|\nabla f\|_2^2} \right\}$$

$$\leq e^{\frac{\pi^2}{8} \lambda^2 L^2}$$

$$= e^{\frac{1}{2} \lambda^2 \sigma^2}$$

$$\sigma^2 = \frac{\pi^2}{4} L^2$$

$$\Rightarrow f(\underline{x}) \sim \text{Sub G} \left(\sqrt{\frac{\pi^2}{4} L^2} \right)$$

$$\Rightarrow \mathbb{P}(|f(x) - \mathbb{E}(f(x))| \geq t) \leq 2e^{-\frac{t^2}{2\pi^2 L^2/4}} \\ = 2e^{-\left(\frac{2}{\pi^2}\right) \frac{t^2}{L^2}} \\ \downarrow \\ 1/2 \\ \square$$

Proof .

$$\mathbb{E} \left\{ \Phi \left(f(x) - \mathbb{E}(f(x)) \right) \right\}$$

$$\varphi(\mathbb{E}(x)) = \Phi(x - \mathbb{E}(x))$$

$$= \mathbb{E} \left\{ \Phi \left(f(x) - \mathbb{E}(f(y)) \right) \right\}$$

$$\leq \mathbb{E}_x \left\{ \mathbb{E}_y \left\{ \Phi \left(f(x) - f(y) \right) \right\} \right\}$$

$$\theta \in [0, \pi/2]$$

$$Z_k = X_k \sin \theta + Y_k \cos \theta$$

$$\theta = 0 \quad Z_k = Y_k \quad f(x) - f(y) = \int_0^{\pi/2} \underbrace{\frac{d}{d\theta} f(z_\theta)}_{\langle \nabla f, z' \rangle} d\theta$$

$$\theta = \pi/2 \quad Z_k = X_k$$

$$\leq \mathbb{E}_{x,y} \left\{ \Phi \left(\frac{\pi}{2} \int_0^{\pi/2} \frac{1}{\left(\frac{\pi}{2}\right)} \langle \nabla f, z' \rangle d\theta \right) \right\}$$

$$= \mathbb{E}_{x,y} \int_0^{\pi/2} \frac{1}{\frac{\pi}{2}} \left[\Phi \left(\underbrace{\langle \nabla f, z' \rangle}_{\nabla f(z)} \right) \right] d\theta$$

$$Z'_k = X_k \cos \theta - Y_k \sin \theta$$

$$\begin{bmatrix} Z_k \\ Z'_k \end{bmatrix} = \begin{bmatrix} \sin \theta & \cos \theta \\ \cos \theta & -\sin \theta \end{bmatrix} \begin{bmatrix} X_k \\ Y_k \end{bmatrix}$$

$$= \{ \phi(\frac{n}{i} < \forall f, \gamma) \}$$